Signals and Systems

Grade Test: Quiz 4

Test Information

Instructions

QUESTION 1: CALCULATED FORMULA 5 **out of 5 points**

The signal 9cos(20πt+7) is applied to the input of a filter whose transfer function is given by $2e^{j3f}$. If the output signal can be written under the for Acos(20πt+B), what is the value of the constant A?

Given Answer: 18

Correct Answer: 18

QUESTION 2: CALCULATED FORMULA

5 out of 5 points

The signal 8cos(20πt+8) is applied to the input of a filter whose transfer function is given by $3e^{j4f}$. If the output signal can be written under the for Acos(20πt+B), what is the value of the constant B?

Given Answer: 48

Correct Answer: 48

QUESTION 3: CALCULATED FORMULA

10 out of 10 points

The signal 2sinc(95t) is sampled at 4 times the Nyquist rate. What is the value of the sampling frequency (in Hz)?

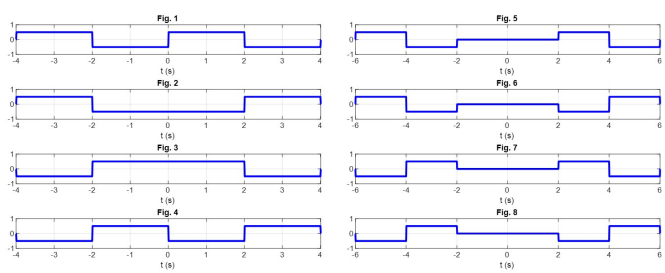
Given Answer: 380

Correct Answer: 380

QUESTION 4: MULTIPLE CHOICE

10 out of 10 points

Which one of the below figures corresponds to the derivative of the signal $-\mathrm{tri}\left(\frac{t+4}{2}\right)+\mathrm{tri}\left(\frac{t-4}{2}\right)$?



Given Answer: Fig. 7

Correct Answer: Fig. 7

QUESTION 5: CALCULATED FORMULA

5 out of 5 points

Consider the signal $x(t)=\text{sinc}(5t)e^{j2\pi 20t}$. If the spectrum of x(t) extends over the frequency range a<f<b, find the value of b.

Given Answer: 22.5

Correct Answer: 22.5

QUESTION 6: CALCULATED FORMULA 10 **out of 10 points**

Consider the signals $x(t)=\text{sinc}^2(6t)$ and $y(t)=\text{sinc}(6t)\cos(2\pi f_c t)$. What is the minimum value of the frequency f_c (in Hz) so that the signals x(t) and y(t) can be reconstructed from the signal x(t)+y(t) with no distortions?

Given Answer: 9

Correct Answer: 9.0

QUESTION 7: CALCULATED FORMULA 5 **out of 5 points**

Consider a signal x(t) whose Fourier transform is given by δ(f-143). If the Fourier transform of the signal x(2t) can be written as δ(f-a), what is the value of a?

Given Answer: 286

Correct Answer: 286

QUESTION 8: TRUE/FALSE 5 **out of 5 points**

The system whose transfer function is given by $H(f)=\dfrac{1}{1+4f^2}$ is real.

Given Answer: True

Correct Answer: True

QUESTION 9: CALCULATED FORMULA 5 **out of 5 points**

5-bit analog-to-digital conversion is applied on a signal that is sampled at 58 Hz. Find the buffer size (in bits) required to store 2 minutes of the signal.

Given Answer: 34,800

Correct Answer: 34,800

QUESTION 10: CALCULATED FORMULA 10 **out of 10 points**

Let x(t)=2sin(π4t)/πt and y(t)=2sin(π10t)/πt. If the signal x(t)*y(t) can be written as Asinc(Bt), what is the value of the constant A?

Given Answer: 16

Correct Answer: 16

QUESTION 11: CALCULATED FORMULA 5 **out of 5 points**

Consider the integral $I = \int_{-\infty}^{+\infty} \mathrm{sinc}\left(\frac{t-b}{a}\right) \mathrm{d}t$. What is the value of I if a=4 and b=15?

Given Answer: 4

Correct Answer: 4

QUESTION 12: MULTIPLE CHOICE 15 **out of 15 points**

The signal $x(t) = 10\mathrm{sinc}(10t)$ is applied to the input of a filter whose transfer function is given by $H(f) = \mathrm{tri}\left(\frac{f}{15}\right)$. What is the expression of the output signal?

Given Answer: $y(t) = \frac{5}{3}\mathrm{sinc}^2(5t) + \frac{20}{3}\mathrm{sinc}(10t)$

Correct Answer: $y(t) = \frac{5}{3}\mathrm{sinc}^2(5t) + \frac{20}{3}\mathrm{sinc}(10t)$

QUESTION 13: TRUE/FALSE | 5 | out of 5 points

The encoding process is not invertible.

Given Answer: False

Correct Answer: False

QUESTION 14: CALCULATED FORMULA | 10 | out of 10 points

The Fourier transform of the signal $x(t) = e^{-at}u(t) + e^{at}u(-t)$ can be written as $X(f) = \frac{b}{a^2 + 4\pi^2 f^2}$. What is the value of b if a=9? (hint: the Fourier transform of the signal w(-t) is W(-f)).

Given Answer: 18

Correct Answer: 18

QUESTION 15: CALCULATED FORMULA | 10 | out of 10 points

What is the energy of the signal 7sinc(t/6)?

Given Answer: 294

Correct Answer: 294

QUESTION 16: CALCULATED FORMULA | 10 | out of 10 points

Consider the quantization of a sampled signal under a fixed dynamic range. If the number of bits per quantization interval increases from 3 to 9, then the mean square error (MSE) will decrease by a factor of:

Given Answer: 4,096

Correct Answer: 4,096

FEEDBACK AND NOTES FOR ATTEMPT

Feedback to Learner

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0

Grading Notes

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0